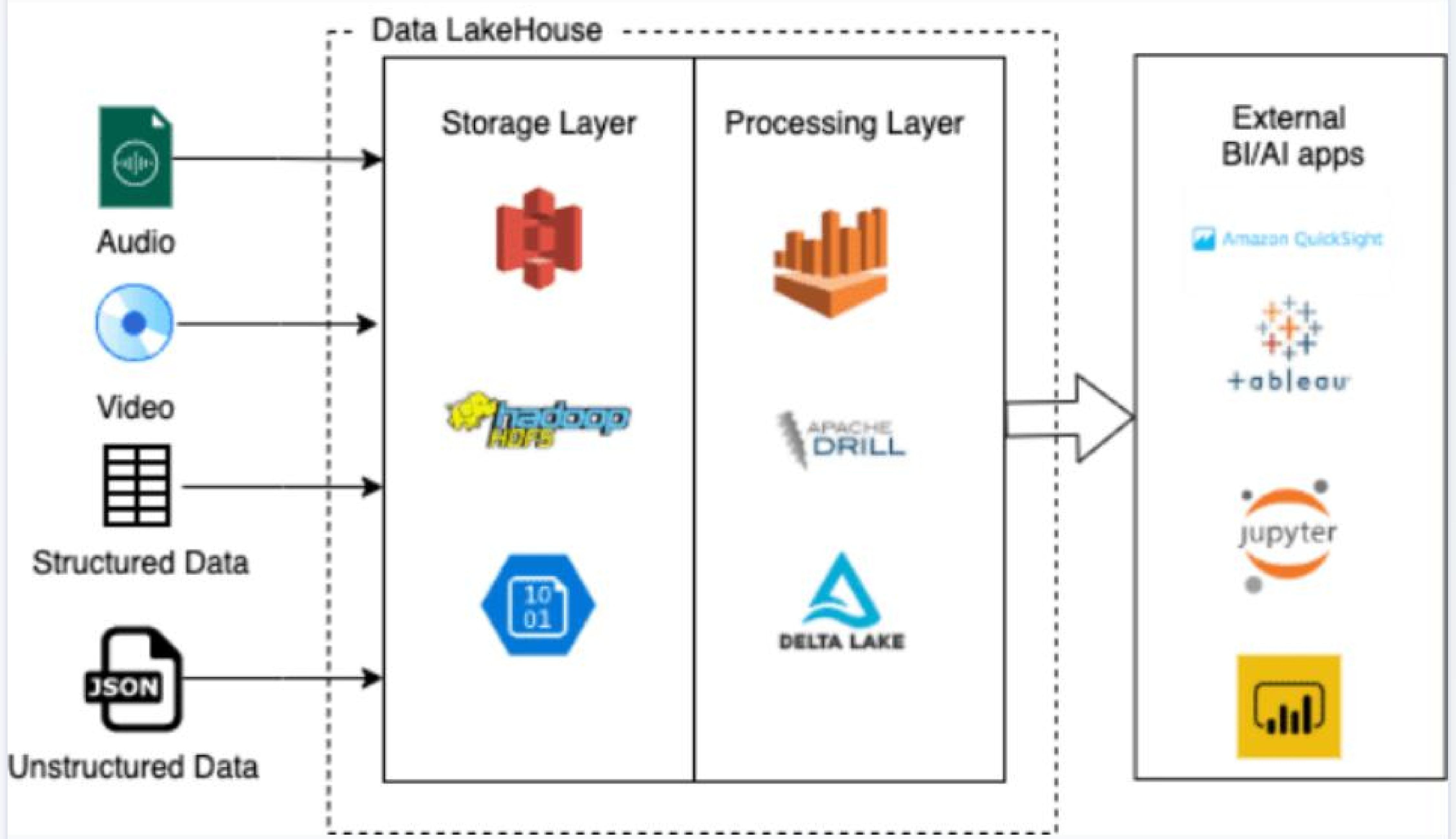
# Microsoft Fabric End-to-End Project

Production-Grade Data Engineering Blueprint inside OneLake

# E2E Platform Architecture

- **Data Sources:** Secure connectivity across transactional SQL platforms, cloud databases, APIs, and raw flat files.
- **OneLake SaaS Lakehouse:** Eliminates data silos, enabling standard Delta Parquet ingestion across computing systems.
- **Unified Consumption:** Pure T-SQL Data Warehouse analytics unified perfectly alongside direct-to-file Power BI connections.



# OneLake: Single Source Of Truth

## Logical Storage Virtualization

Acts as an internal system for all workspaces. Built directly on Azure ADLS Gen2, offering a single logical directory system and schema layer for the entire global enterprise, regardless of multi-tenant workspaces.

## Zero-Copy Shortcuts

Enables zero-movement connections across multiple data clouds (Azure Storage, AWS S3, Google Cloud). Access files seamlessly without triggering massive data movement tasks or network sync routines.

# The Medallion Structure

### Bronze (Raw Staging)

Ingests unmodified raw files. Preserves native structures, history indexes, and acts as the point of recovery.

### Silver (Cleaned)

Applies cleansing rules, filters nulls, enforces schemas, deduplicates keys, and creates queryable datasets.

### Gold (Analytical Facts)

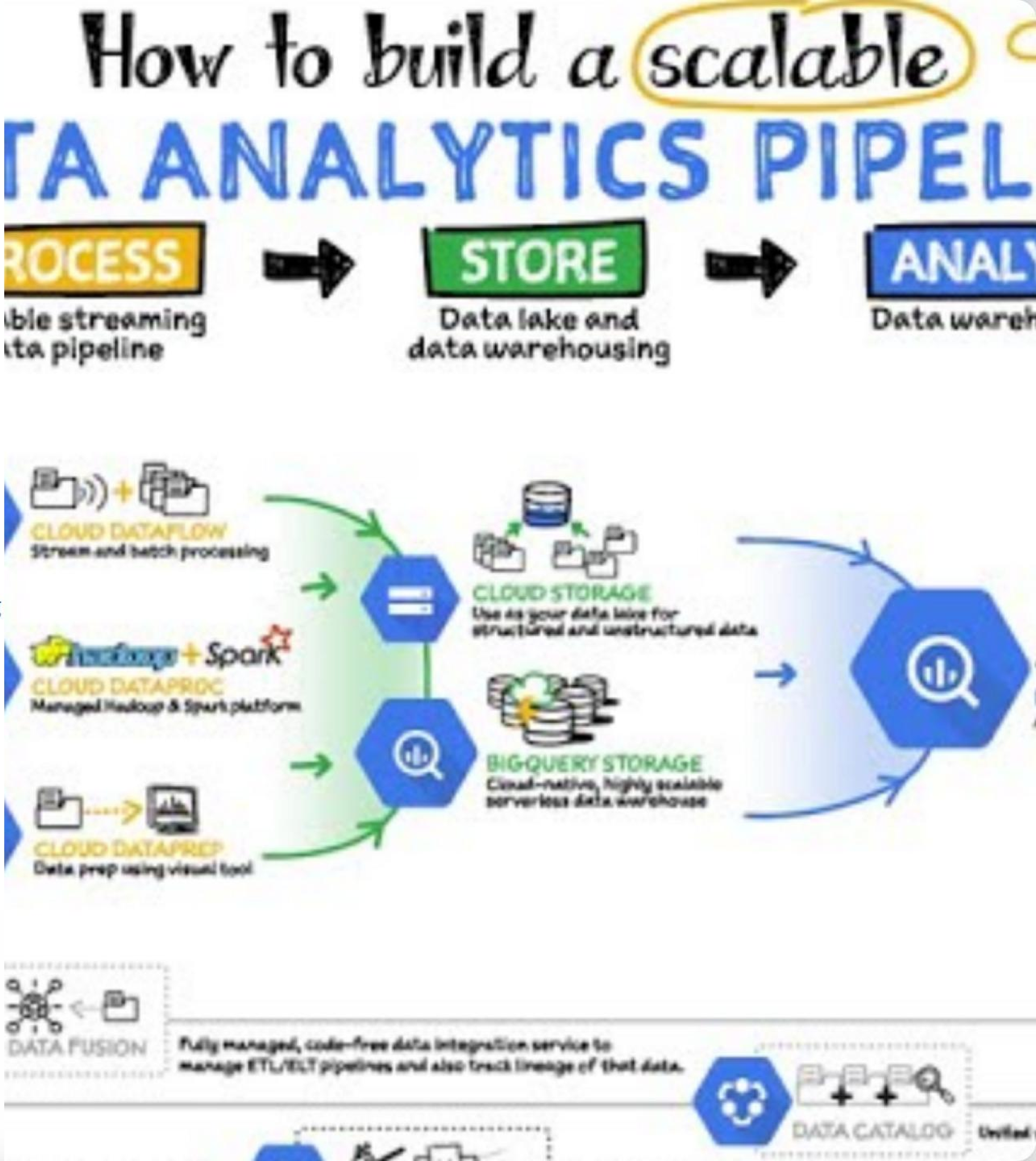
Assembles Kimball Star Schemas. Fact & Dim structures calculated ready for executive-level analytical models.

## Ingestion Engine

### Automated Data Factory

Build powerful pipeline networks scaling across local databases and cloud APIs using Fabric's serverless orchestration loops.

Applies metadata-driven variables to dynamically scale loops across 50+ relational operational schemas within a single copy pattern.



# Transformation Tooling

## Enterprise PySpark

Utilized for high-performance runs and complex business transformations. Employs Microsoft V-Order write algorithms to automatically compact and optimize raw Delta file write-performance.

## Dataflows Gen2

Empowers business intelligence analysts and engineers with an intuitive Power Query visual interface, mapping source schemas to OneLake storage with zero local coding overhead.

# Incremental Loading Framework

| Strategy Pattern | Resource Overhead | Complexity | Key Fabric Tooling Used |
| --- | --- | --- | --- |
| **Full Table Flush** | High (Full DB Scan) | Ultra-Low | Pipeline Copy Activity |
| **Watermarking (Delta Date)** | Medium (Filter Queries) | Medium | Pipeline Lookup + PySpark |
| **Change Data Capture (CDC)** | Low (Logs parsing) | High | Fabric Mirroring / Synapse SQL |

# Fabric SQL Warehousing

## Dedicated SQL Warehouse

Designed for relational enterprise databases. Provides complete T-SQL capabilities including transactions, DDL/DML support, stored procedures, and views designed for standard SQL developers.

## Lakehouse Endpoint

A read-only SQL virtualization endpoint layer built over Lakehouse Delta tables. Enables analysts to instantly query parquet files using T-SQL without duplicating files to a separate database.

# Power BI DirectLake Connectivity

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Data Refreshes Required

## Bypassing Import & DirectQuery

DirectLake reads Delta files directly from OneLake storage, loading data in-memory on demand without hitting SQL Server or data warehouses. It offers import-level rendering speeds with DirectQuery data freshness.

# Access & Enterprise Governance

- **Granular RBAC Security:** Establish segregation of analytical access through Workspace Roles (Admin, Member, Contributor, and Viewer) to govern platform developer modifications.
- **SQL Row-Level Security:** Restrict dataset access on semantic models directly using dynamic Row-Level Security (RLS) linked to Entra ID user identities.
- **Purview Lineage Tracking:** Native integration with Microsoft Purview tracks data flow continuously from external sources, through tables, down to downstream report visual boxes.

# Platform Performance Gains


*Performance benchmarking highlights that Microsoft Fabric Direct Lake reduces data latency by up to 95%, while PySpark optimizations shorten ETL schedules significantly.*

# Questions & Answers

Let's discuss implementing your Microsoft Fabric Data Platform migration journey.

**Enterprise Data Platform Architecture Team**

# Image Sources

https://sonra.io/wp-content/uploads/2021/12/word-image-2.png

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